Patrick Thangarajah

Simulation & FEA | Testing & Validation | Mechanical Design | Prototyping

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Summary

Mechanical Engineering MASc graduate with experience in finite element modeling, model validation, impact mechanics, and dynamic simulation.

Skills

Finite Element Analysis	LS-DYNA, LS-PrePost, nonlinear FEA, material modeling, contact, boundary conditions
Design & Prototyping	SolidWorks, Onshape, Engineering Drawings, 3D Printing, Lathe, Mill, Bandsaw, Drill Press
Project Engineering & Validation	Testing & validation, DOE, troubleshooting, process improvement, technical problem-solving
Programming and Data Analysis	MATLAB, Python, JMP, DOE, Microsoft Excel, C/C++

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Research in impact mechanics, finite element analysis, and model validation
- Graduate coursework average: 90%

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*
- Biomechanics and Medical Devices Option

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Validated a THOR-5F ATD FE model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Investigated material response, contact interactions, boundary conditions, and model behavior to understand discrepancies between simulation predictions and experimental measurements
- Designed and analyzed a 104-simulation DOE-based FE parametric study evaluating the effect of seat belt restraint anchor locations on rear seat occupant response, identifying a rearward and inboard position that reduced peak chest deflection by 16% below the study average
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

R&D Engineer (Part-time) — *Remote*
Echoscope — *May 2023 - Sep 2023*
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functional testing; all five were approved and used by the client
- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*
- Created standardized work instructions and SOPs verified against engineering drawings and customer specifications
- Reduced fiberglass cutting waste by 50% through process analysis, and proposed a material handling workflow projected to halve pick-and-place time

Engineering Intern — *Victoria, BC, Canada*
Breathe Global — *Jul 2021 - Aug 2021*
- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

Projects

Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Jan 2024 - Apr 2024*
- Verified 5 polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics
- Determined that all 5 foams met the regulatory head injury limit but exceeded the regulatory neck injury limit at high speed, and that the softest foam reduced head injury risk by roughly 18–20% relative to the stiffest

Soft Distensible Bladder Model - Capstone Project — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2022 - Apr 2023*
- Validated pressure chamber performance using the integrated sensor and control system, demonstrating stable, repeatable pressurization at 2.2 kPa (13.8s ramp-up) up to a maximum internal bladder pressure of 4.75 kPa

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- Watson, B., Seeburrun, T., **Thangarajah, P.**, Tang, K., Tylko, S., & Cronin, D. S. (2026). *Experimental and numerical investigation of belt loading for small-stature female occupants in rear seat frontal crash*. Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV), Toronto, ON, Canada, May 12–15, 2026.